ISHA PATRO

ip2529@nyu.edu | (347) 241-6603 | Linkedin | Portfolio | Github | Blogs

EDUCATION

NEW YORK UNIVERSITY, TANDON SCHOOL OF ENGINEERING Brooklyn, NY
Master of Science in **Financial Engineering** Expected 05/27
GPA: 3.5/4.0

VELLORE INSTITUTE OF TECHNOLOGY, AP Andhra Pradesh, India
Bachelor in **Computer Science** 07/21
GPA: 3.6/4.0

PROGRAMMING & TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C++, R, VBA
Web Development: React, RESTful APIs, AG Grid, Streamlit, Dash, Tablue
Data Science: Statistical Modeling, Machine Learning, Deep Learning, LLMs, RAG, AI Agents, Causal Inference
Databases & Cloud: SQL, Neo4j, q/kdb+, AWS, MongoDB, Microsoft Excel, Bloomberg Terminal
Quantitative Finance: Alpha Mining, Financial Modeling, Risk Management, Fixed Income Analytics, Time Series Forecasting

EXPERIENCE

J.P MORGAN CHASE & CO. 01/21 - 08/25
Software Developer Engineer - II, MBS Division - Fixed Income Pricing Team (Banglore, India)
- Led the development and post-launch ownership of Trade Activity Monitoring (TAM) as Epic Owner, collaborating with stakeholders to achieve a 50% latency reduction and significantly improve trading floor execution efficiency.
- Pioneered and architected a comprehensive Mortgage Matrix modernization initiative driving 12% latency improvement supporting $5B+ annual trading volume.
- Designed and implemented Trade Reporting and Compliance Engine (TRACE) in an agile environment collaborating with cross-functional teams to boost data processing efficiency by 30% and streamline regulatory reporting workflows.

RESEARCH / OPEN SOURCE CONTRIBUTION

FinOKF: Typed Data Format and Certification Protocol for Trustworthy LLM Financial Reasoning 06/25 - Present
- Architecting a typed profile of the Open Knowledge Format with a certification layer whose soundness is mechanically proven in Lean 4, re-executing LLM-generated financial calculations in exact rational arithmetic, then stress-testing the system against 100+ adversarial attacks on real SEC EDGAR filing data.

Narrative Risk Allocation for Commodity Portfolios 01/26 - Present
- Engineering an end-to-end differentiable deep-learning portfolio allocator that embeds 125K+ financial news articles into volatility-spillover signals across 9 commodity futures, achieving a 1.73 Sharpe ratio over a 16-year walk-forward backtest.

Implied Prepayment Surfaces via Physics-Informed Neural Networks in Agency MBS Markets 11/25 - 03/26
Under review at Journal of Computational Finance.
- Built a PDE-constrained neural network that extracts implied prepayment from agency MBS prices and statistically outperforms the option-adjusted spread benchmark out-of-sample, validated with leakage-safe backtesting and deflated performance statistics.

qutePandas (Open Source Contribution) 06/25 - Present
- Published a high-performance Pandas-compatible execution layer backed by kdb+/q, achieving order-of-magnitude speedups(10x–100x+) on large-scale analytical workloads while preserving Python-first research ergonomics.

HONORS / AWARDS / COMPETITIONS

- Awarded the prestigious Numerix Women in Finance Scholarship, a **$20,000 honor**, 2025
- Two-time 1st Place Winner, Engineering Clinics Expo, VIT **($1,200)** - across all engineering disciplines, 2017-2018

ACADEMIC PROJECTS

VORTEX 05/26
- Architected a full-stack quantitative risk analytics platform: Gaussian HMM regime detection, 5-method VaR backtesting, XGBoost volatility forecasting, and a LangGraph multi-agent LLM system.

Monte Carlo Pricing of European Barrier Options 10/25
- Developed a Monte Carlo engine under risk-neutral GBM to price path-dependent barrier options (Down-and-Out, Knock-In/Out), incorporating barrier monitoring and sensitivity analysis.

Advanced NASDAQ Market Prediction System with Deep Learning Optimization 08/25
- Engineered a sophisticated LSTM neural network system for high-frequency NASDAQ market prediction, achieving 26.7% improvement in forecasting accuracy through advanced hyperparameter optimization.